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Fiber Polytopes, and Beyond

The *secondary polytopes* of Gel'fand, Kapranov & Zelevinsky [231] are a quite recent development that grew out of their theory of “ $\mathcal{A}$ -hypergeometric functions.” After Izrail M. Gel'fand and Andrei Zelevinsky presented the miraculous construction at the 1989 Symposium on Combinatorics and Geometry in Stockholm, a lot of effort was put into understanding what is going on geometrically. It seems that the definitive answer — unexpectedly simple — is the one supplied by Louis Billera and Bernd Sturmfels [78, 79, 534], who introduced the *fiber polytope* of a projection of polytopes and showed that secondary polytopes arise in the special case where we project the $(n - 1)$ -simplex to a given polytope with n vertices.

The main goal in this lecture is to develop geometric intuition for the fiber polytope construction. Many interesting examples have been studied, and among them we will concentrate on the construction of the permutahedron and the associahedron as fiber polytopes. With “fiber polytopes intuition” we then construct the permuto-associahedra: nice new polytopes proposed by Kapranov [313] as combinatorial objects and realized as polytopes in Reiner & Ziegler [453].

So fiber polytopes help to solve special cases of a difficult general problem: the *construction of polytopes with specified combinatorics*. Using the Lawrence construction (Lecture 6), one can see that the solution for any specified face lattice is difficult: this “algorithmic Steinitz problem” is as difficult as the solution of general polynomial systems over the reals; see [96, p. 407]. (The inverse problem, the complete description of the face lattice of a given polytope, is not trivial either: see Lecture 1 and Exercise 9.0).